
MI 102: Manufacturing Techniques

Course Code	MI 102
Course Name	Manufacturing Techniques
L-T-P-C	2-0-2-3
Prerequisites	NIL Sem. Both
Approval	5th Senate (Oct 2013)

Course Contents

- **Introduction** : Engineering materials, their manufacturability and applications;
- **Casting** : Pattern materials, pattern types, allowances, molding sand, composition and properties, cores, casting defects and their remedies, plastic parts molding;
- **Machining**: Lathe, drilling, milling and grinding machines and their operations, cutting tools used;
- **Joining** : Welding fundamentals, types of welded joints, types of welding processes, gas welding process, manual metal arc welding, welding defects and remedies, Soldering and brazing, their applications in electronics industry;
- **Forming**: Forging, rolling, extrusion, wire drawing and tube drawing, sheet metal operations, forging defects and remedies;
- **Advance Manufacturing Process**: Introduction to advanced manufacturing technique and their application.

References

1. De Garmo P. E., Black J.T., Kohser R. A., **Materials and Processes in Manufacturing**, Prentice Hall of India Pvt. Limited.
2. Kalpakjian S., Schmid S. R., **Manufacturing Engineering and Technology**, Pearson Education.
3. Groover M. P., **Fundamentals of Modern Manufacturing**, John Wiley and Sons Inc.
4. Lindberg R. A., **Processes and Materials of Manufacture**, Prentice Hall India Limited.